

Port Congestion Classification

Machine Learning & Neural Network Project Report

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1. Introduction

The goal of this project was to analyze port congestion data and predict congestion levels using machine learning and neural network models. Predicting congestion levels can help organizations improve logistics planning, reduce delays, and optimize supply chain operations.

2. Dataset Description

The dataset used in this project was port_congestion.csv. The congestion index was converted into three categories: Low Congestion, Medium Congestion, and High Congestion. The dataset contains numerical and categorical features related to port and shipping activities. Categorical variables were encoded before model training and the data was split into training and testing sets.

3. Machine Learning Models

Three machine learning models were evaluated: Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN). The evaluation metrics included Accuracy, Precision, Recall, and F1-score.

4. Neural Network

A neural network was implemented using TensorFlow and Keras. The architecture included Dense layers with 128 and 64 neurons, Dropout layers for regularization, and a Softmax output layer. The model was trained for 15 epochs using the Adam optimizer.

5. Results & Comparison

Random Forest achieved the highest performance among all tested models and was selected as the final model for this project.

6. What I Learned

This project provided practical experience in data preprocessing, feature engineering, machine learning model evaluation, and neural network development. It demonstrated the importance of selecting the appropriate model for real-world supply chain applications.

Model Performance Summary

Model	F1-Score
Logistic Regression	0.945
Random Forest	0.960
KNN	0.922
Neural Network	0.932

Conclusion

Based on the experimental results, Random Forest was selected as the final model because it achieved the highest F1-score of 0.960. The project demonstrates that machine learning techniques can effectively classify port congestion levels and support better decision-making in supply chain management.